

SMART EXPENSE SPLITTER AND GROUP BUDGET MANAGEMENT SYSTEM

Team No: 3

Challa Amrutha	2520030412
Dhurga Meenakshi Yenduri	2520030270
Earla Ashwith Reddy	2520030470

Guide: Dr. Y. Subbarayudu

| PROJECT OVERVIEW

The Smart Expense Splitter and Group Budget Management System is a web-based application developed to simplify the management of shared expenses among friends, roommates, families, and project teams. The system enables users to create groups, record expenses, track budgets, and manage settlements efficiently. By automating calculations and providing financial insights, the application reduces manual effort and improves transparency in group financial management.



Problem Statement

Managing shared expenses manually becomes difficult when multiple people contribute different amounts toward group activities. Traditional methods such as spreadsheets or calculators are prone to calculation errors and make it difficult to track payments and budgets. Existing expense-splitting applications mainly focus on recording expenses and dividing bills but provide limited support for budget forecasting, spending analysis, and intelligent financial recommendations.

| OBJECTIVES

➡ The primary objective of this project is to develop a smart and user-friendly platform for managing group expenses and budgets.

➡ The system aims to automate expense calculations, provide real-time budget monitoring, predict budget overruns, analyze spending patterns, and identify unusual expenses.

➡ Another important goal is to minimize unnecessary transactions between members and help users make better financial decisions through meaningful spending insights.

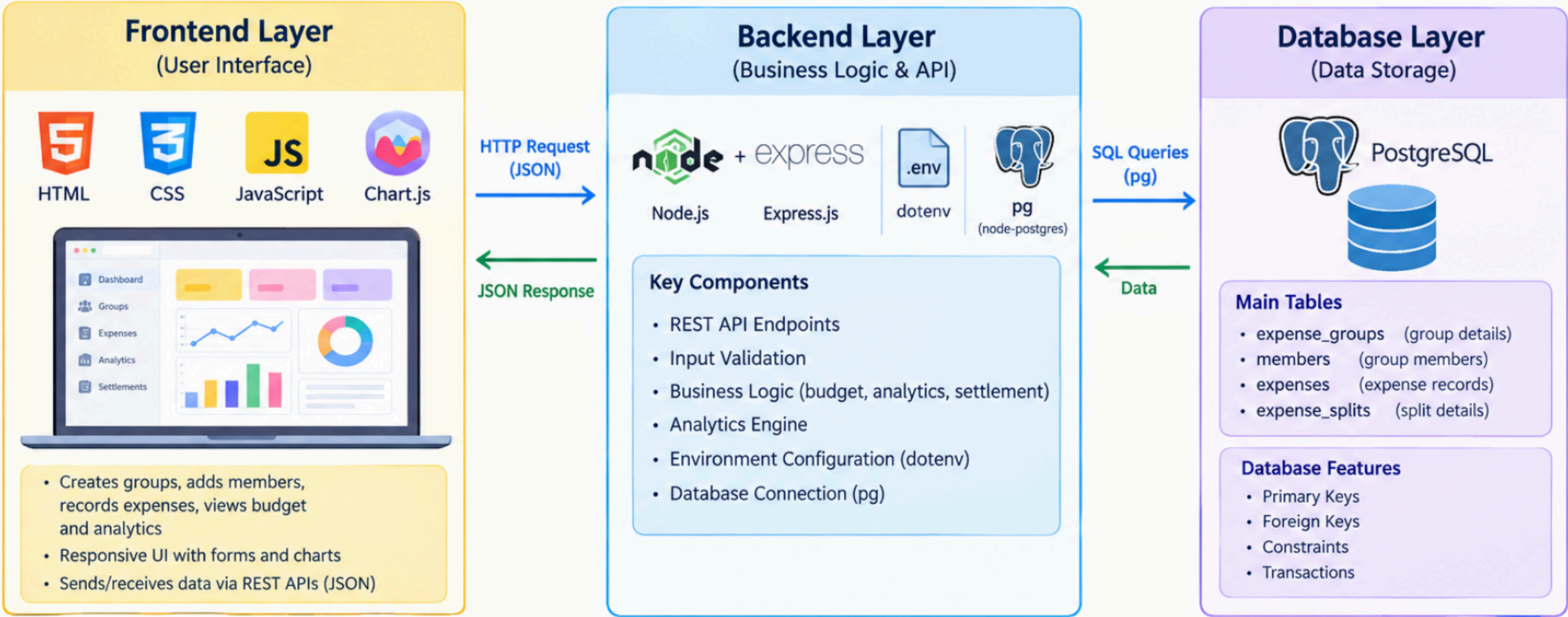
PROPOSED SOLUTION

The proposed system provides an intelligent solution for group financial management through automated expense splitting and budget tracking. It includes features such as predictive budget alerts, spending-pattern analysis, optimized debt settlement, and what-if budget simulations. These capabilities allow users to monitor expenses effectively, avoid budget overruns, and settle debts with fewer transactions, making the overall process more transparent and efficient.

System Architecture

System Architecture

Smart Expense Splitter and Group Budget Management System



How It Works

- 1 User performs an action on the frontend (e.g., add expense)
- 2 Frontend sends REST API request to backend (JSON)
- 3 Backend validates, processes business logic and interacts with database
- 4 Database stores/retrieves data and returns response
- 5 Frontend updates UI and charts with the latest information

Tools & Technologies



Node.js

express

Express.js



PostgreSQL



VS Code



Git & GitHub

npm

npm



dotenv



Homebrew
(macOS)

MODULES

The project is divided into several functional modules to ensure organized and efficient operation. The User and Group Management Module handles authentication and group creation. The Expense Management Module records and categorizes expenses, while the Budget Management Module tracks spending against predefined budgets. Additional modules such as Settlement Management and Smart Analytics provide debt optimization, spending analysis, and personalized financial insights for users.

TECHNOLOGY STACK

Frontend: HTML5, CSS3, JavaScript, Chart.js

Backend: Node.js, Express.js

Database: PostgreSQL

Database Connectivity: pg (node-postgres)

Configuration Management: dotenv

API Architecture: REST API

Data Exchange Format: JSON

Development Environment: Visual Studio Code

Package Manager: npm

Version Control: Git & GitHub

Database Installation Tool: Homebrew (macOS)

Server Environment: Node.js Runtime

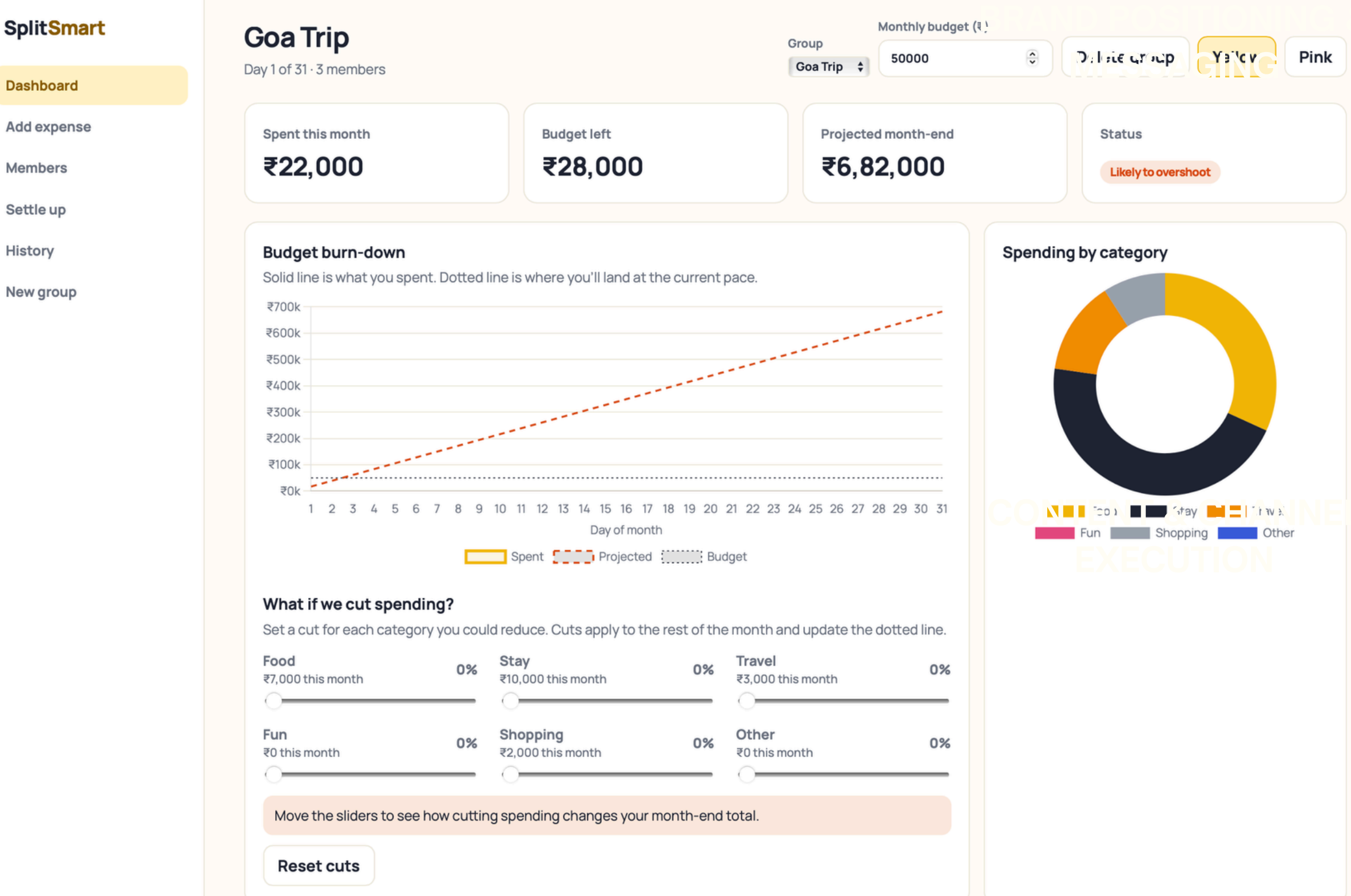
Testing & API Validation: Postman

Platform: Web-Based Application

— DATABASE DESIGN

The database is designed to efficiently manage and organize all information related to groups, members, expenses, budgets, and settlements. PostgreSQL is used as the relational database management system, ensuring data integrity through primary keys, foreign keys, constraints, and transactions. The database structure enables accurate expense calculations, secure storage of financial records, and efficient retrieval of information. The system consists of four major tables: `expense_groups`, which stores group details and budgets; `members`, which stores member information linked to groups; `expenses`, which records all expense transactions; and `expense_splits`, which maintains expense-sharing details among members. This structured design ensures consistency, scalability, and reliable management of group financial data.

Implementation and Results



The Smart Expense Splitter and Group Budget Management System was successfully implemented using a full-stack architecture. The frontend provides an interactive dashboard where users can create groups, add members, record expenses, and monitor budgets. The backend processes requests, validates user input, performs calculations, and communicates with the database. Features such as automatic expense splitting, balance calculation, budget tracking, settlement generation, and spending analytics were successfully integrated. Testing demonstrated that the system accurately calculates individual shares, updates balances in real time, and generates meaningful financial insights. The developed prototype effectively meets the objectives of simplifying expense management and improving financial transparency.

Implementation and Results

SplitSmart

Dashboard

Add expense

Members

Settle up

History

New group

Members

Name	Paid	Share	Balance
Ash	₹7,000	₹7,333	owes ₹333 Remove
Riya	₹13,000	₹7,333	gets back ₹5,667 Remove
Sunny	₹2,000	₹7,333	owes ₹5,333 Remove

Add a member

Name

Add member

Expenses

Clear all expenses

Expenses far above their category's average are flagged.

Date	Expense	Category	Paid by	Split	Amount	
2026-10-03	Bus from the beach	Travel	Riya	Equal	₹3,000	Delete
2026-10-02	Keychains	Shopping	Sunny	Equal	₹2,000	Delete
2026-10-01	Dinner	Food	Ash	Equal	₹7,000	Delete
2026-10-01	Hotel	Stay	Riya	Equal	₹10,000	Delete

Create a group

Group name

Monthly budget (₹)

Members (comma separated)

Create group

Implementation and Results

SplitSmart

Dashboard

Add expense

Members

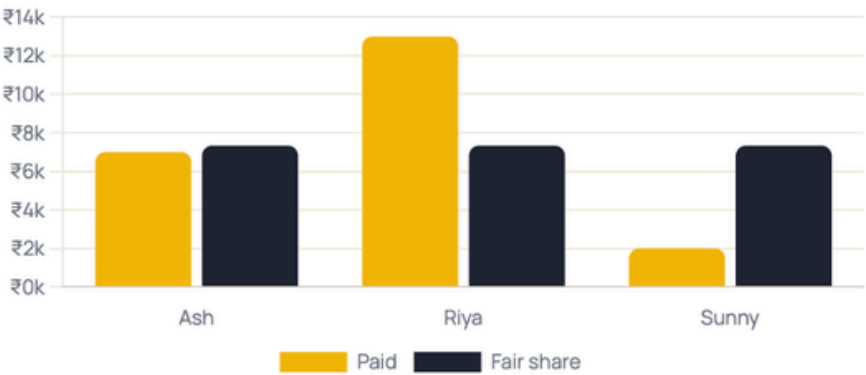
Settle up

History

New group

Paid vs fair share

Who covered more than their share.



Settle up

The fewest payments that clear every balance.

Sunny pays Riya	₹5,333
Ash pays Riya	₹333

Add expense

What was it for

Amount (₹)

Category

Paid by

Date

Split

Dinner at Fisherman's

Food

Ash

10 / 01 / 2026

Equally

☒ Ash

☒ Riya

☒ Sunny

Save expense

Members

Name	Paid	Share	Balance
Ash	₹7,000	₹7,333	owes ₹333 Remove
Riya	₹13,000	₹7,333	gets back ₹5,667 Remove
Sunny	₹2,000	₹7,333	owes ₹5,333 Remove

Add a member

Name

Add member

FUTURE SCOPE

The project can be further enhanced by integrating advanced technologies and additional features to improve user experience and functionality. Future improvements may include AI-powered financial recommendations that help users optimize spending habits, OCR-based receipt scanning for automatic expense entry, and voice-based expense recording for greater convenience. Mobile application support can increase accessibility, while multi-currency management and payment gateway integration can make the system suitable for international and real-world financial scenarios. Advanced analytics and predictive models can also be incorporated to provide more accurate budgeting insights and spending forecasts.

CONCLUSION

The Smart Expense Splitter and Group Budget Management System successfully addresses the challenges associated with managing shared expenses and group budgets. By automating expense calculations, tracking budgets, analyzing spending patterns, and optimizing debt settlements, the system reduces manual effort and minimizes errors. The application promotes transparency among group members and provides valuable financial insights that support informed decision-making.

Overall, the project demonstrates how modern web technologies and database systems can be combined to create an efficient, reliable, and user-friendly solution for collaborative financial management.

THE END

